

Lecture_4_743

Group_ID: 743

Hand exercises

The covariance matrix of a dataset has the following eigenvalues:

$$\lambda_1 = 5, \lambda_2 = 3, \lambda_3 = 1, \lambda_4 = 0.5$$

with the corresponding eigenvectors

$$v_1, v_2, v_3, v_4$$

1. Which eigenvector becomes the first principal component?

The first principal component is the largest eigen value thus $\lambda_1 = 5$

2. What percentage of the total variance is retained if only the first principal component is kept?

$$\begin{aligned} var_p &= \frac{\lambda_1}{\sum_{i=1}^4 \lambda_i} \\ &= \frac{5}{5 + 3 + 1 + 0.5} \\ &= 0.5263 \end{aligned}$$

3. How many principal components are needed to retain at least 90% of the total variance?

Initially the total value of λ for retaining at least 90% is found

$$\begin{aligned} solve\left(\frac{x}{\sum_{i=1}^4 \lambda_i} = 0.9, x\right) \\ x = 8.55 \end{aligned}$$

Thus $\sum_{i=1}^n \lambda_i$ must be $\lceil x \rceil$ or larger

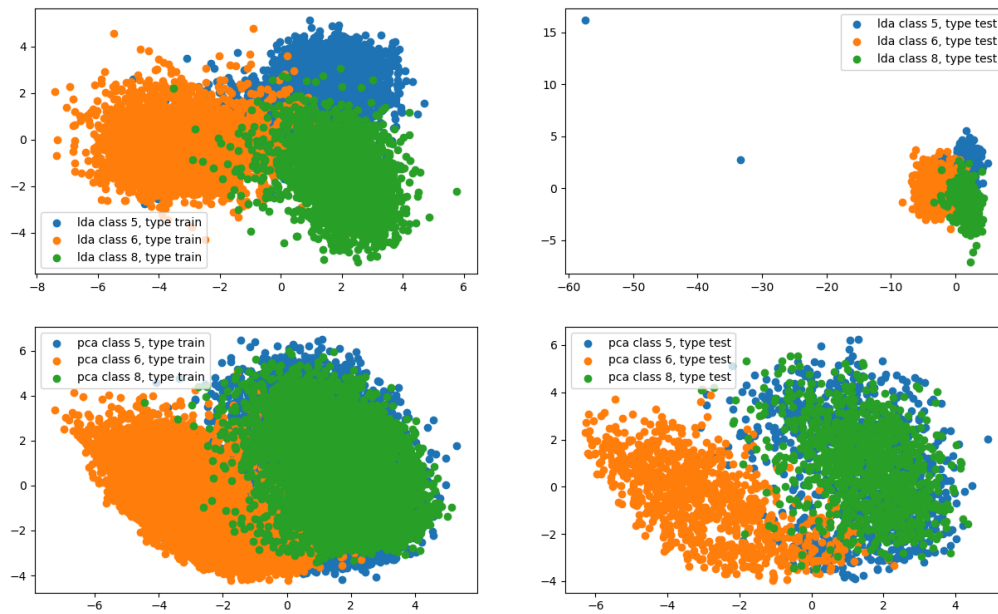
$$\sum_{i=1}^3 \lambda_i = 5 + 3 + 1 = 9$$

thus $n = 3$ meaning that for retaining at least 90% of the total variance 3 principal values are needed.

Coding Exercises

training set of 60,000 examples, and a test set of 10,000 examples; each having a size of $28 \times 28 = 784$ (concatenating the pixels).

Choose 3 classes (5, 6, 8) and transform / reduce the dimensions down to 2 dimensions using LDA and PCA and compare accuracies and error rate of classification



When comparing the PCA and LDA it shows that LDA is better at maximizing the differences of each class, as class 5 and 8 is very close in PCA compared to LDA.

This is also carried into classification where LDA classification has a more accurate classification rate of 0.944 compared to PCA's accuracy of 0.710